

CS 313: Computer Networks

Course Code	CS 313
Course Name	Computer Networks
L-T-P-C	3-0-2-4
Intended for	BTech CSE (Third year)
Approval	42nd Senate (Jan 2024), in the B.Tech CSE curriculum; 54th BoA (May 2024)

Course Contents

- **Introduction** (4 Hours):
 - What is the Internet? Services and protocols.
 - Core and Edge Network, Packet-switched and circuit-switched networks
 - Layered architecture, Data Encapsulation, the end-to-end argument, history of the Internet.
 - [Lab] Networking tools in Linux: ping, traceroute, etc. Introduction packet analyzers such as tcpdump, Wireshark.
- **Application Layer** (4 Hours):
 - Services of the application layer, network applications, where is the application layer?
 - Application layer protocols, Hypertext Transfer Protocol, Email and Simple Mail Transfer Protocol (SMTP), Secure Shell (SSH), Domain Name System (DNS), File Transfer Protocol (FTP).
 - [Lab] Socket programming.
- **Transport Layer** (7 Hours):
 - Transport layer services and overview, Principles of reliable data transfer, UDP and TCP, Flow control and congestion control.
 - TCP congestion control, TCP implementation: Slow start, congestion avoidance, fast retransmit and fast recovery, Evolution of TCP, legacy TCP and latest variants.
 - [Lab] Introduction to network simulators: ns2/ns3. Simulation-based study of TCP and UDP.
- **Network Layer** (9 Hours):
 - Network layer services and overview, architecture of a router, Switching and Forwarding, Datagram Approach (connectionless Service), Virtual Circuit Approach (connection-oriented Service).
 - The Internet Protocol, Addressing and NAT, IPv4, IPv6 and Dynamic Host Configuration Protocol (DHCP).
 - Unicast and multicast routing, Routing algorithms: Link-state routing and Distance-vector routing.
 - Interaction of transport and network layer. Role of network layer in congestion control.
 - Software-Defined Networking (SDN) and its evolution, Network management and associated protocols. [Lab] Study of routing algorithms using ns2/ns3 or network emulation using Mininet.
- **Link Layer** (8 Hours):
 - Link layer services, overview, link layer addressing, Address Resolution Protocol (ARP).
 - Error detection and correction, Cyclic Redundancy Check (CRC) and Checksum.
 - Multiple access protocols: channel partitioning, random access (ALOHA, CSMA and its variants), taking turns. Link-layer addressing, Ethernet and its evolution, Switches and VLANs.
 - [Lab] Virtual networking in Linux.
- **Wireless and next-gen networks** (4 Hours):
 - WiFi and 802.11 standard.
 - Cellular Internet Access, Routing and handoffs in mobile networks.
 - Edge computing, architecture for cloud networking and IoT. [Lab] Simulation-/measurement-based study of wireless networks.
- **Advanced topics** (6 Hours):
 - Firewalls, Proxy, VoIP, protocols for video streaming, network economics, Content Delivery Networks.

Text Books

1. JF. Kurose and KW. Ross, **Computer Networking: A top-down approach**, 8th Edition, Pearson, 2022.

References

1. L. Peterson and B. Davie, **Computer Networks: A Systems Approach**, 6th Edition, Morgan Kaufmann Publishers.
2. BA. Forouzan, **Data Communications and Networking**, 5th Edition, McGraw Hill.
3. MJ. Donahoo and KL. Calvert, **TCP/IP Sockets in C: Practical Guide for Programmers**, Morgan Kaufmann Publishers.